The Inception of In-Context Learning: Minimal Training Structure for Contextual Candidate Promotion

Paper 0.8 — Controlled ICL Mechanism Research Line

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Abstract

We ask for the smallest training structure and Transformer in which context promotes a target that is low-ranked without demonstrations to rank one. A leakage-audited D0–D4 ladder excludes the critical $C \to 3$ mapping and compares correct demonstrations with shuffled, wrong-chain, irrelevant, reversal, and no-context controls. We first classify the computational tasks: the present D4 is finite permutation constraint completion (T3), not rule induction. Under its uniform k-symbol permutation prior, all-but-one demonstrations reduce $H(Y \mid X) = \log_2 k$ to $H(Y \mid X, D_{\text{ctx}}) = 0$, so the context supplies $\log_2 k$ bits and the oracle ceiling is one. A 54-cell coarse sweep finds 16 individual positives, including a one-layer, width-16, one-head model, but none of five Pareto-small candidates passes all three seeds. Thus D4 identifies a seed-sensitive acquisition frontier, not yet a stable general ICL mechanism. In parallel, contextual-copy calibration recovers a known microscopic circuit in one competent two-layer checkpoint: early pair binding, later associated-value transport, and rank-one promotion principally in second-layer attention. Value patching from shuffled context reduces accuracy from 100% to 20.8%, while demonstration-value ablation reduces it to 37.5%. Fresh replications fail the 95% three-seed gate, so this circuit is a validated local calibration rather than a stable minimality claim. The combined result separates information sufficiency, behavioral acquisition, and the physical mechanism available after acquisition.

1 Question

Let Y be a candidate with low context-free rank, $\operatorname{rank}(Y \mid X) \gg 1$. We seek the smallest training distribution $\mathcal{D}^\star$ and smallest model $M^\star$ for which $\operatorname{rank}(Y \mid D_{\text{ctx}}, X) = 1$, while Y remains low-ranked without D_{ctx}. Define

$$G_{\text{ICL}}(Y) = m(Y \mid D_{\text{ctx}}, X) - m(Y \mid X), \quad m(Y) = z(Y) - \max_{y \neq Y} z(y).$$

The threshold is crossed when contextual margin becomes positive on held-out relations and the effect disappears under matched shuffled/wrong-context controls.

2 Minimal motivating example

Train local orders $A \to B, B \to C$ and $1 \to 2, 2 \to 3$, but never train $C \to 3$ or the complete paired mapping. Test

$$A\,1, \qquad B\,2, \qquad C\,?$$

and ask whether context aligns the two local chains strongly enough to promote 3.

3 Minimal-dataset ladder

D0: random/no useful structure.
D1: isolated local successor chains only.
D2: many isomorphic independent chains.
D3: paired alignment or selection from a finite transformation family.
D4: episode-specific permutation completion preventing global mapping memorization.
D5: identifiable parametric or structured rules, introduced only after the simpler regimes are separated.

At each stage add exactly one structural ingredient.

3.1 Computational taxonomy and information sufficiency

We distinguish in-weight association (T0), direct copy (T1), associative/induction copy (T2), finite constraint completion (T3), finite task selection (T4), parametric rule induction (T5), compositional rule induction (T6), and procedural or recursive induction (T7). These labels describe the computation licensed by the generator, not what a trained network necessarily implements. Table 1 locates the present ladder. D1–D2 test whether local relational exposure creates contextual alignment pressure; the current paired D3 is a finite alignment/completion test. Proposed D3-B/C variants would instead select among a known finite mapping or transformation set and must separate seen from unseen mappings.

Table 1: Dataset taxonomy. "Determined" means that context plus the declared prior fixes the answer; it does not imply that a model learned the computation.

Regime	Type	Computational task	Target shown	Determined	Iterative
D0–D2	T0	association/local structure	no	no	no
D3-A	T3	partial alignment completion	no	yes	no
D3-B/C	T4	finite task selection	no	yes	no
D4	T3	permutation constraint completion	no	yes	no
D5 unary	T5	parametric modular rule	no	gated	no
D5 composed	T6/T7	compositional/procedural rule	no	gated	possibly

For D4, let π be uniform over the $k!$ permutations and let the query identify one key. Before demonstrations, its value is uniform, hence

$$H(Y \mid X) = \log_2 k.$$

The prompt shows $k-1$ key–value pairs. The omitted value—which is not explicitly present in the context—is uniquely fixed by permutation complement, giving

$$H(Y \mid X, D_{\mathrm{ctx}}) = 0, \qquad I(Y; D_{\mathrm{ctx}} \mid X) = \log_2 k,$$

and Bayes/oracle accuracy one. In the experiments $k = 3$: the mapping space has $3! = 6$ members, each prompt contains two demonstrations, and the context contributes $\log_2 3 \simeq 1.585$ bits. Consequently, a selective D4 positive establishes genuine contextual dependence, but specifically for T3 elimination/completion; it is not evidence of a newly inferred function.

3.2 D5 validity gate

We implemented a CPU-only exact-posterior validator for successor, add-n, multiply-by-a, affine, and square functions on $\mathbb{Z}_k$, for $k \in \{5, 7, 11\}$. Each episode enumerates the declared finite hypothesis family, filters parameters consistent with the demonstrations, computes the induced answer distribution at the query, and rejects the episode unless that answer has zero entropy. It also stores the number of consistent hypotheses and the minimum identifying prefix. Across 360 validation episodes (24 per family/modulus cell), all passed with posterior answer entropy zero; 284 required one demonstration and 76 required two. The latter comprise all 72 affine episodes and four multiply-by-a episodes. This verifies generator admissibility only. No D5 Transformer has been trained, so the paper offers no behavioral or mechanistic evidence for T5–T7 rule induction.

4 Dataset and model phase diagrams

Search over number of relation families, examples per family, task/mapping entropy, chain length, symbol reuse, template variation, and position variation. Start with

$$L \in \{1, 2, 3, 4, 6\}, \qquad d_{\mathrm{model}} \in \{8, 16, 32, 64\},$$

and one or two heads where feasible. The goal is a behavioral phase boundary, not maximum benchmark performance.

5 Controls

Every positive cell compares no demonstrations, correct demonstrations, shuffled outputs, wrong-chain demonstrations, same-length irrelevant context, order perturbations, random initialization, held-out symbols, and held-out relation families. Audit that the critical direct mapping never occurs in training and that the target remains low-ranked without context.

6 Layerwise mechanism

At every boundary record residual state, SA update, FF update, target logit, rank, margin, top-k candidates, and entropy. Compare context-free and context-conditioned trajectories. The first question is where Y becomes a serious candidate; the second is where it becomes rank one.

7 Mechanistic hypotheses

Test whether attention aligns corresponding relational roles, transports a successor/order state, or directly increases support for the target. Test whether FF already encodes reusable successor transformations, maps an attention-transported relation into the target direction, or simply sharpens a candidate. These are competing hypotheses.

8 Causal experiments

Ablate heads; zero or replace SA/FF updates; patch states from correct, shuffled, wrong, role-matched, and matched-norm controls; block individual demonstration tokens; and remove one local relation from training and retrain. Prefer causal loss of contextual promotion over attention-map interpretation.

9 Before/after acquisition

Find closely matched $\mathcal{D}^-$ and $\mathcal{D}^+$ conditions immediately below and above the ICL threshold. Compare head utility, SA/FF updates, residual target projection, relation-role similarity, cross-layer covariance, and weights. This threshold comparison is central.

10 Minimality

After finding a positive regime, delete one training family, relation type, positional regularity, symbol overlap, layer, head, or width. If the effect survives, simplify again.

11 Interpretation

Do not assume the Transformer is performing regression, Bayesian inference, or implicit gradient descent. Those may be useful behavioral approximations at larger scale. Prefer the smallest measured mechanism consistent with the evidence:

$$\boxed{\text{learned local relations} + \text{contextual alignment/transport} + \text{candidate promotion.}}$$

12 D0–D4 behavioral results

The reference L2/W32/H2 model was evaluated with three seeds. D0 never passes. D1 and D2 reach 87.5% correct-context accuracy, but their controls can perform equally well; local order is therefore insufficient. D3 and D4 reach 100% correct-context accuracy, yet only one of three seeds passes the complete context-dependence gate. Across the full 54-cell coarse surface, 16 cells pass individually.

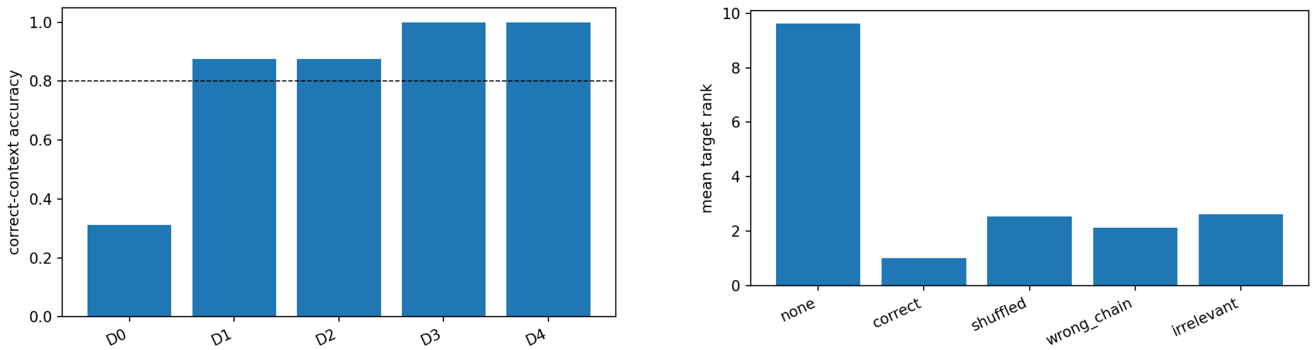

Figure 1: Left: dataset-ladder accuracy at the reference architecture. Right: D4 seed-11 rank promotion and matched controls. High correct-context accuracy is insufficient when a control also succeeds or the context-free target is already favored.

The smallest local positive is L1/W16/H1: 100% correct context, zero context-free accuracy, mean context-free rank 26.5, zero negative-control accuracy, and margin gain 18.37. Replication rejects it: seeds 23 and 37 achieve only 25%. Four further Pareto candidates also fail the three-seed gate. L2/W16/H1 and L6/W8/H1 pass two seeds; all others pass one. The supported result is a sharp, non-monotonic, seed-sensitive acquisition frontier. D4 tracing is consequently blocked rather than selected from a favorable seed.

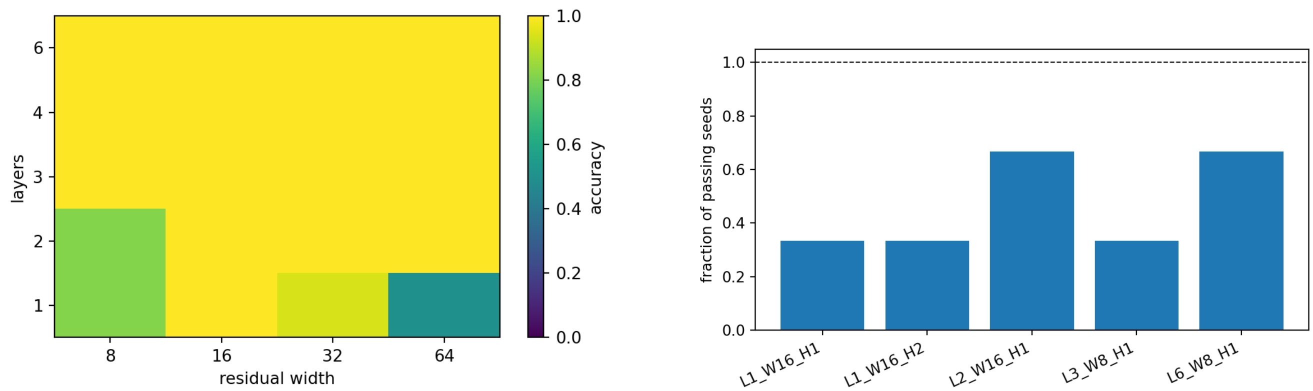

Figure 2: Coarse seed-11 architecture surface (left) and three-seed replication fraction for Pareto-small positives (right). Deeper or wider models are not monotonically more selective.

13 Contextual-copy calibration

The C2 calibration uses balanced episode-specific associations, so a fixed key-to-value map cannot solve the query. One L2/W64/H2 checkpoint passes at 100%, context-free rank 3.0, and 14.6% maximum association-control accuracy. Exact Q/K/V tracing finds layer-0 head-0 previous-token mass 0.441. At layer 1, the two heads place 0.624 and 0.664 attention probability on the associated demonstration value, while repeated-key attention is only 0.039 and 0.068. This is a pair-binding then value-routing circuit, not a claim that attention probability itself is mass flow.

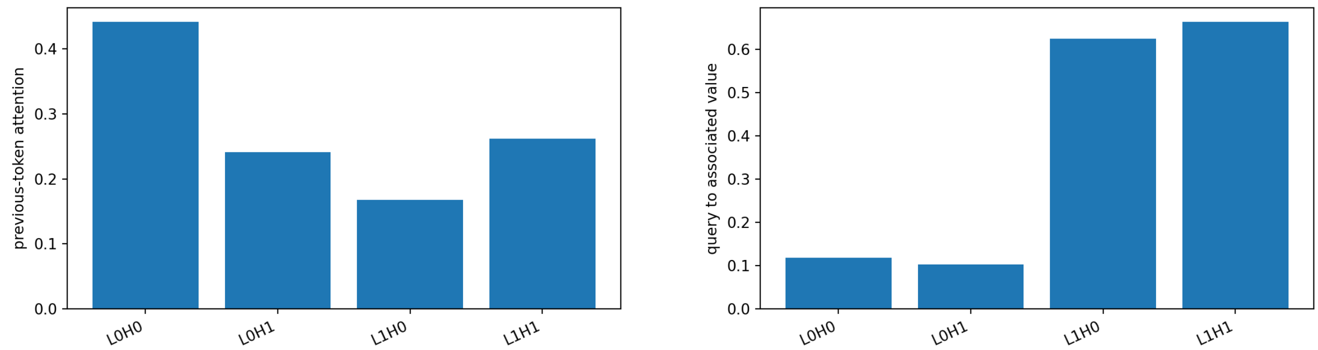

Figure 3: Role-aligned copy calibration: early previous-token binding and later query-to-associated-value routing.

Causal tests agree. Layer-1 demonstration-value ablation lowers top-1 accuracy from 100% to 37.5% and damages margin by 15.24. Patching all layer-1 values from shuffled context lowers accuracy to 20.8% (margin damage 24.89); patching keys retains 77.1%. Layer-0 head-0 ablation lowers accuracy to 54.2%, while both layer-1 heads are useful. Mean margin progresses from −28.33 initially through −21.98 after layer-0 SA and −3.83 after layer-0 FF, then to +15.54 after layer-1 SA. Final FF preserves rank one but slightly reduces margin to +13.65. Thus later attention supplies the decisive target support.

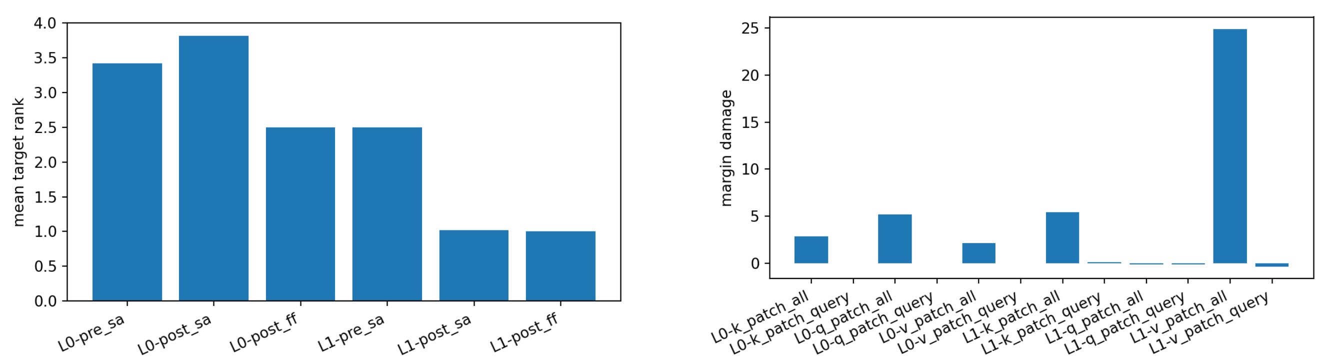

Figure 4: Target trajectory (left) and causal Q/K/V patch damage (right). The target becomes rank one principally in second-layer attention, and shuffled value transport is most destructive.

This is a local instrumentation calibration. Fresh L2/W64/H2 replications at 256 episodes reach 89.6%, 70.8%, and 66.7%, all below the 95% gate; L2/W128/H2 reaches only 41.7%, 54.2%, and 33.3%. Increasing episode diversity to (E=1024) makes seeds 11 and 23 competent at both the nominal and doubled update budgets (100% each), but seed 37 reaches only 52.1% and 81.2%, respectively. A targeted fourfold-budget run does not rescue seed 37: accuracy falls to 60.4%. The non-monotonic trajectory rules out the simple explanation that this failure merely needed more updates. It instead supports an initialization- and optimization-sensitive acquisition boundary. Hence the circuit is physically validated where acquired, while reliable three-seed acquisition remains unestablished.

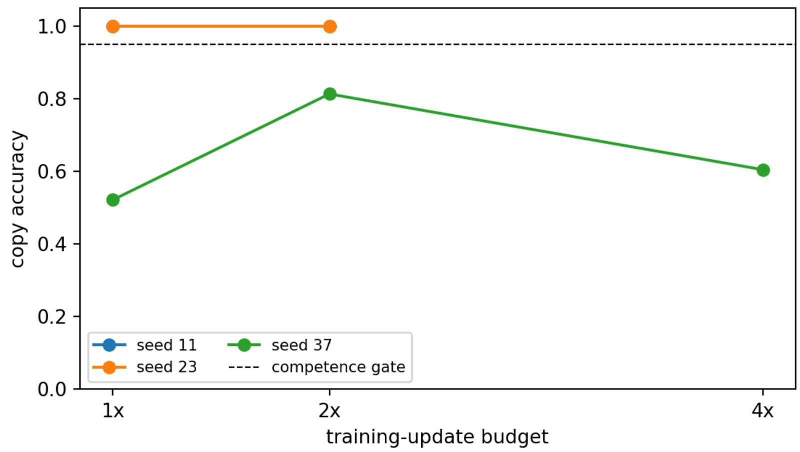

Figure 5: Copy acquisition at (E=1024) versus update budget. Seeds 11 and 23 pass at the measured 1x and 2x budgets; seed 37 improves at 2x but regresses at 4x. The isolated 4x point is a targeted failure diagnostic, not a matched three-seed estimate.

14 Answers and limitations

Local order alone is not sufficient. Partial/fresh mapping pressure creates local positives, but no tested D4 architecture is seed-stable. D4 is nevertheless information-sufficient: its context determines the held-out value exactly. The failure is therefore acquisition instability rather than dataset ambiguity. Conversely, exact D5 validation makes those proposed episodes admissible but supplies no model evidence that parametric, compositional, or procedural induction has been learned. Where contextual copy is acquired, the smallest measured mechanism is *bind pairs, match the query role, transport the associated value, promote its logit*; second-layer SA is decisive and FF is supportive rather than the final promoter. Increasing copy diversity rescues two seeds, but update-budget scaling is non-monotonic for the remaining seed and does not close the replication gate. We find no need for regression, Bayesian, or implicit-gradient metaphors. We do not yet claim a stable smallest copy or D4 model, nor trace D4 from a selectively chosen seed. The next experiment should alter acquisition reliability—optimizer or curriculum, rather than update count alone—while keeping the leakage, information-sufficiency, and control gates fixed.

Table 2: Final evidence audit. A behavioral positive is not promoted to a stable mechanism unless its stated replication gate passes.

Component	Evidence	Gate	Supported interpretation
D0–D4 generator	exact audits and controls	pass	valid bounded task ladder
D4 behavior	16/54 seed-local positives	0/5 architectures stable	acquisition frontier only
Copy behavior	competent traced checkpoint	fresh three-seed gate fails	local calibration
Copy mechanism	tracing plus causal patches	pass in that checkpoint	pair binding/value routing
D5	360 exact-posterior episodes	generator only	admissible future tasks

15 Reproducibility

Generators and phase runners are under `src/cl/semantic/` and `src/cl/experiments/`. The shared tracer and analysis modules use the `paper08_*` prefix. Configurations are under `configs/paper08/`. The taxonomy audit is configured by `taxonomy_v1.json`; its exact D5 posterior rows and machine-readable D0–D5 classification are under `results/taxonomy/`. Tracked artifacts include raw phase/control rows, five three-seed D4 replications, copy replication and exact trace tables, the (E=1024) training-budget diagnostic, compressed Q/K/V matrices, 18 figures, and summaries. Dense checkpoints remain locally reproducible and git-ignored.

16 Relation to Papers 0.5–0.7

Paper 0.5 studies predictive refinement once evidence is accessible. Paper 0.6 studies learned semantic/compositional relations. Paper 0.8 asks how context combines learned relations to create a new prediction without weight updates. Paper 0.7 now supplies validated formal-task infrastructure, while its learned formal-model frontier remains pending; no Paper 0.7 result is used as evidence here.

17 Scaling policy

Do not scale merely to improve performance. Scale only if the minimal model cannot express the capability, a mechanistic hypothesis requires additional components, or external validity becomes the explicit question.

18 Contribution boundary

The present contribution is not a stable minimal ICL model. It is a disciplined boundary: an information-valid constraint-completion task whose observed positives remain seed-sensitive, together with a contextual-copy checkpoint in which causal interventions locate the binding and value-routing mechanism. A future positive would need to identify the smallest statistically sufficient environment and architecture across seeds, then give a causal account of where signed target-logit support arises. D5 is intentionally not launched in this audit.